

TSFS12 Hand-in 4: Collaborative Control

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1 Objective

This report addresses the exercises in TSFS12 Hand-in Exercise 4: Collaborative Control. The goal is to implement and analyze formation control strategies presented in Lecture 8 and the provided assignment sheet. The exercises treat three families of formation control laws: position-based, displacement-based, and distance-based formation control; followed by an extra assignment on measurement errors and directed distance control. The original exercise sheet used for this submission is included as the hand-in reference.

2 Exercises

2.1 5.1 Position-Based Formation Control

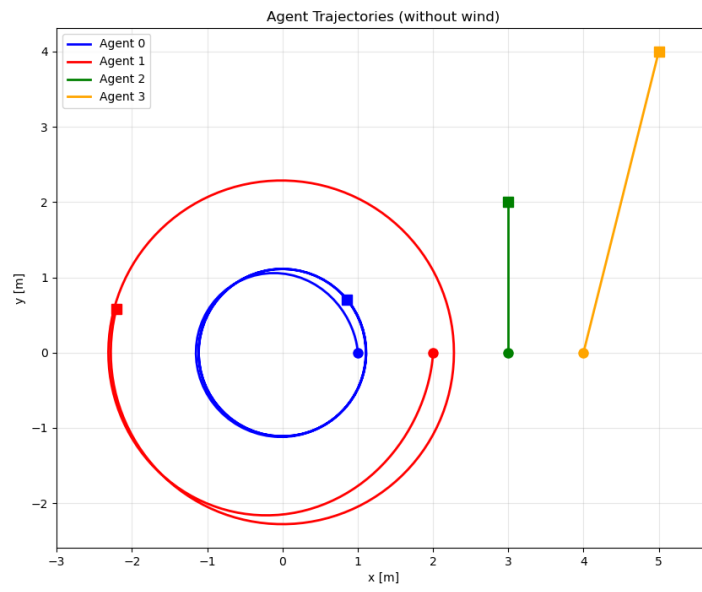
Problem statement. Replace the single-integrator model by a double-integrator: $\ddot{p}_i = u_i$ and use the controller: $u_i = k_v(v_i^* - v_i) + k_p(p_i^* - p_i)$. Include a constant wind force in the x -direction and simulate the model with and without wind.

Model and controller. The agent state is defined as: $x_i = (p_i \ v_i)^\top$, $\dot{x}_i = (v_i \ u_i + w)^\top$ where $w = (w_x, 0)^\top$ represents a constant acceleration from wind.

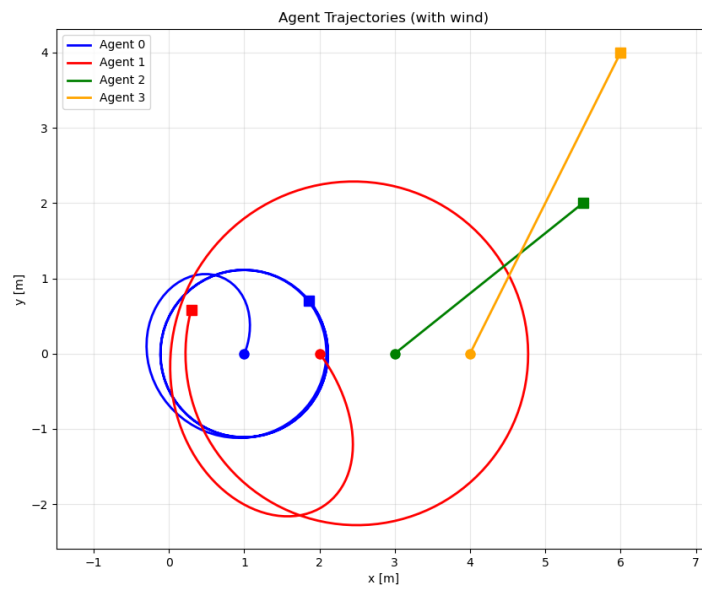
The control law becomes: $u_i = k_v(v_i^* - v_i) + k_p(p_i^* - p_i)$

Closed-loop dynamics. Let $\tilde{p}_i = p_i - p_i^*$ and $\tilde{v}_i = v_i - v_i^*$. Neglecting reference acceleration $\ddot{p}_i^*$, the error dynamics are: $\dot{\tilde{p}}_i = \tilde{v}_i$, $\dot{\tilde{v}}_i = -k_v\tilde{v}_i - k_p\tilde{p}_i$. The characteristic polynomial for each coordinate is $\lambda^2 + k_v\lambda + k_p = 0$. Stability is ensured when $k_v, k_p > 0$.

Effect of constant wind. With a constant disturbance w , the steady-state offset satisfies: $-k_p\tilde{p}_i + w = 0 \Rightarrow \tilde{p}_i = \frac{w}{k_p}$. Hence, larger k_p reduces steady-state position error.

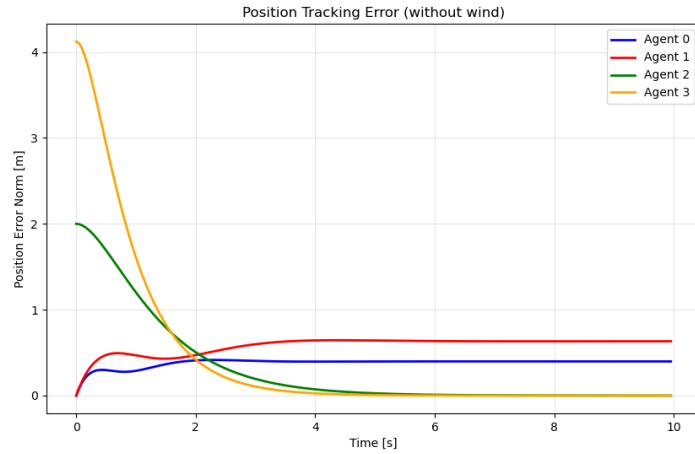


(a) no wind

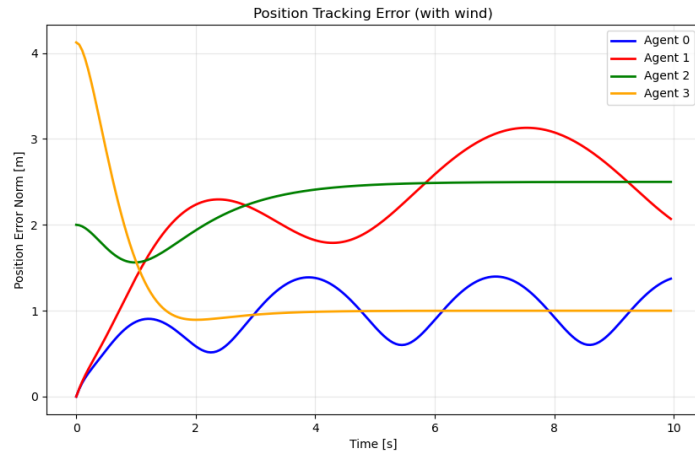


(b) wind

(c) Agent trajectories without wind and with wind .



(a) no wind



(b) wind

(c) Agent position error without wind and with wind .

Plots

Implementation notes.

- Extend the single-integrator function to a double-integrator including wind input.
- Implement a PD controller on (p_i, v_i) .
- Simulate both cases (with and without wind) and compare positions and velocities.

2.2 5.2 Displacement-Based Formation Control

Problem statement. Eight agents, modeled as double-integrators $\ddot{p}_i = u_i$, form a vertical string with spacing 1. Agent 1 (top) measures absolute position and aims for $(0, 4)$; other agents use relative measurements to neighbors.

Control laws. $u_1 = k_v(v_1^* - v_1) + k_p(p_1^* - p_1)$ $u_i = k_p \sum_{j \in \mathcal{N}_i} (p_j - p_i - p_j^* + p_i^*) + k_v \sum_{j \in \mathcal{N}_i} (v_j - v_i - v_j^* + v_i^*)$

(1) Simulation ex 5.2.1 For $k_p = k_v = 2$, initial $p_i = (i, 0)$, $v_i = 0$, and desired leader position $(0, 4)$.

Linearized formation dynamics: $\ddot{\tilde{P}} + k_v L \dot{\tilde{P}} + k_p L \tilde{P} = 0$ where L is the Laplacian of the chain graph. Stability holds for $k_p, k_v > 0$.

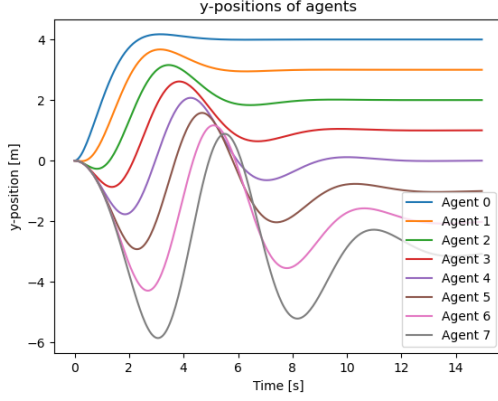


Figure 3: Vertical velocities of agents.

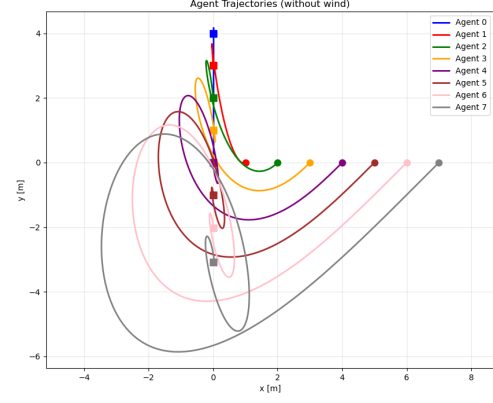
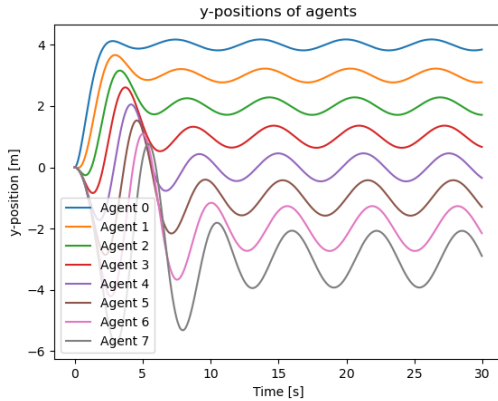
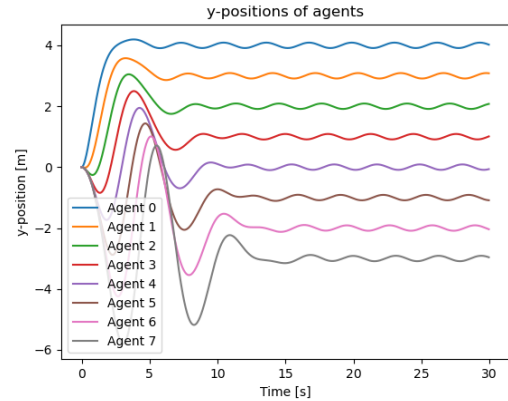


Figure 4: Positions of agents.

(2) Simulation ex 5.2.2 Remove edges from node $(i - 1)$ to node i for $i = 3, \dots, 8$. Add harmonic motion to leader: $p_1^*(t) = (0, 4 + 0.2 \sin(\omega t))$. Simulate for $\omega = 1$ rad/s and $\omega = 2$ rad/s. Lower frequency yields smaller phase lag and smoother tracking.



(a) $\omega = 1$



(b) $\omega = 2$

Figure 5: Vertical velocities of agents for $\omega = 1$ and $\omega = 2$.

(4) Sign convention analysis. The control law for the leader uses $(v^* - v)$ and $(p^* - p)$, while others use $(v_j - v_i - v_j^* + v_i^*)$. Flipping signs inconsistently makes agents react in opposite directions, destabilizing the formation.

2.3 5.3 Distance-Based Formation Control

Problem statement. Desired positions: $(-1, 3), (0, 0), (0, 6), (1, 5), (1, 1), (1, 3), (-1, 1), (-1, 5)$. Single-integrator model: $\dot{p} * i = u_i$. Leader uses $u_i = k_p(p_i^* - p_i)$. Others use gradient

control: $u_i = -\nabla * p_i \varphi_i = k_p \sum_{j \in \mathcal{N}^* i} \gamma' * ij(|p_j - p_i|) \frac{p_j - p_i}{|p_j - p_i|}$ with potential function: $\varphi_i = \frac{k_p}{2} \sum_{j \in \mathcal{N}^* i} \gamma * ij(|p_j - p_i|)$ and $\gamma_{ij}(|p_j - p_i|) = (|p_j - p_i|^2 - |p_j^* - p_i^*|^2)^2$

(1) **ex 5.3.1** Top node moves on a circle of radius 2. For $k_p = 1$ and $k_p = 5$, the distance error decays faster for higher k_p .

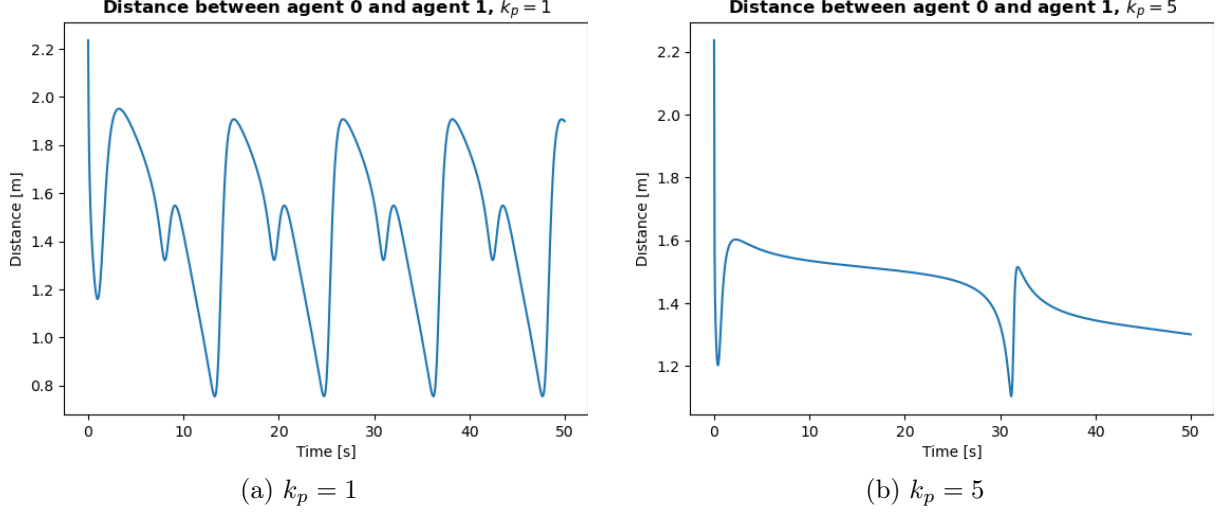


Figure 6: Distance between top node and a neighbor for $k_p = 1$ (solid) and $k_p = 5$ (dashed).

(2) **ex 5.3.2** When the bottom agent also measures absolute position, the formation's orientation is fixed because two pinned nodes remove rotational degrees of freedom.

t = 50.0

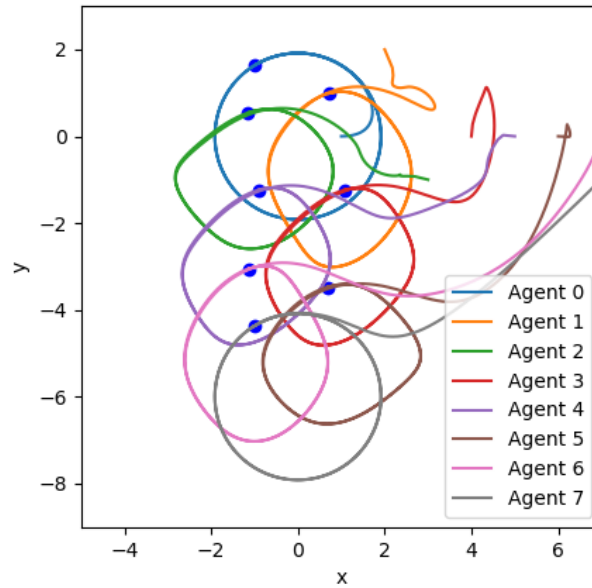


Figure 7: Paths of agents with top and bottom nodes measuring absolute position.

3 Concluding Discussion

This report analyzed and implemented three formation-control methods and two extensions. Main observations:

- Displacement-based control ensures convergence when the graph is connected and leader is pinned.
- Distance-based control enforces pairwise constraints but is sensitive to measurement errors.
- Directed distance control requires graph asymmetry analysis for stability.